Lecture 32 Canonical Correlation II: Sample

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

1. Sample canonical correlations
2. Relationship to Other Correlations
3. Summary and Remark

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Sample canonical correlations

- The sample canonical variables are obtained by applying the population method to $\mathbf{S}$
- one can use $\mathbf{R}$, but then all interpretations are in terms of linear combinations of *standardized* variables.
- To compute: obtain $\mathbf{S}_{11}^{-1/2}$, $\mathbf{S}_{22}^{-1/2}$ and then $\tilde{\boldsymbol{S}}_{12} = \mathbf{S}_{11}^{-1/2}\mathbf{S}_{12}\mathbf{S}_{22}^{-1/2}$.
- Use

$$\mathtt{s} = \mathtt{svd}(\tilde{\boldsymbol{S}}_{12},\ \mathtt{nu} = \mathtt{p},\ \mathtt{nv} = \mathtt{q})$$

to get

$$\mathbf{G} = \mathtt{s\$u}, \mathbf{H} = \mathtt{s\$v} \text{ and } \left(\hat{\rho}_1^*, \cdots, \hat{\rho}_p^*\right) = \mathtt{s\$d}.$$

- Then the normalized rows of $\tilde{\boldsymbol{A}} = \mathbf{G}^T\mathbf{S}_{11}^{-1/2}$ and $\tilde{\boldsymbol{B}} = \mathbf{H}^T\mathbf{S}_{22}^{-1/2}$ are the $\left(\mathbf{a}_i^T, \mathbf{b}_i^T\right)$, and $(U_i, V_i) = \left(\mathbf{a}_i^T\mathbf{x}, \mathbf{b}_i^T\mathbf{y}\right)$ are the canonical pairs.

Example

- Example — see web site.
-
```
> round(cbind(A, B),2)
# coefficients of U and V
[,1]  [,2]  [,3]  [,4]
[1,] -0.95 -0.31 -0.59 -0.80
[2,] -0.54  0.84 -0.77  0.63

> round(1/diag(Atilde%*%t(Atilde)),2) # var of U's
[1] 1.23 0.64
> round(1/diag(Btilde%*%t(Btilde)),2) # var of V's
[1] 1.19 0.80

> # Canonical correlations:
> round(diag(R),2)
[1] 0.74 0.03
```

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Example

- Interpretation:

$$\begin{aligned}(U_1, V_1) &= (-.95X_1 - .31X_2, -.59Y_1 - .80Y_2),\\ (U_2, V_2) &= (-.54X_1 + .84X_2, -.77Y_1 + .63Y_2);\\ |\text{CORR}[U_1, V_1]| &= .74, \quad |\text{CORR}[U_2, V_2]| = .03.\end{aligned}$$

- The variances are

$$\begin{aligned}\text{VAR}[U_1] &= 1.23, \text{VAR}[V_1] = 1.19;\\ \text{VAR}[U_2] &= .64, \ \text{VAR}[V_2] = .80;\end{aligned}$$

with

$$\text{VAR}[U_1] + \text{VAR}[U_2] = 1.87 < 2 = \text{VAR}[X_1] + \text{VAR}[X_2].$$

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Relationship to Other Correlations

- If $\mathbf{\Sigma}_{12} = \mathbf{0}$ then all $\rho_i^* = 0$. We can test this hypothesis; if rejected we might test the hypothesis that only the first k are significantly non-zero. Large sample likelihood ratio tests with Bartlett corrections:

- $H : \mathbf{\Sigma}_{12} = \mathbf{0}$. Reject for large values of

$$-2\log\Lambda \overset{how?}{=} n\log\left(\frac{|\mathbf{S}_{11}|\,|\mathbf{S}_{22}|}{|\mathbf{S}|}\right) = -n\log\prod_{i=1}^{p}\left(1-\hat{\rho}_i^{*2}\right) \xrightarrow{L} \chi^2_{pq}. \tag{2.1}$$

  The Bartlett correction replaces n by $n - 1 - \frac{p+q+1}{2}$.

- $H : \rho_1^* \geq \cdots \geq \rho_k^* > 0 = \rho_{k+1}^* = \cdots = \rho_p^*$:

$$-\left(n - 1 - \frac{p+q+1}{2}\right)\log\prod_{i=k+1}^{p}\left(1-\hat{\rho}_i^{*2}\right) \xrightarrow{L} \chi^2_{(p-k)(q-k)}.$$

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More Examples

- Examples. Two sets of variables — job characteristics and measures of job satisfaction ($p = 5$, $q = 7$):

$$\mathbf{x} = \begin{pmatrix} \text{feedback} \\ \text{task significance} \\ \text{task variety} \\ \text{task identity} \\ \text{autonomy} \end{pmatrix},$$

$$\mathbf{y} = \begin{pmatrix} \text{supervisor satisf'n,} \\ \text{career-future satisf'n} \\ \text{financial satisf'n} \\ \text{workload satisf'n} \\ \text{company identification} \\ \text{kind-of-work satisf'n} \\ \text{general satisf'n} \end{pmatrix}.$$

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More Examples

- See code on web site — the coefficients of the first two canonical pairs are

$$\begin{vmatrix} & \hat{\mathbf{a}}_1 & \hat{\mathbf{a}}_2 \\ X_1 & 0.62 & 0.25 \\ X_2 & 0.29 & -0.48 \\ X_3 & 0.25 & -0.61 \\ X_4 & -0.03 & 0.25 \\ X_5 & 0.68 & 0.52 \end{vmatrix}, \begin{vmatrix} & \hat{\mathbf{b}}_1 & \hat{\mathbf{b}}_2 \\ Y_1 & 0.55 & -0.07 \\ Y_2 & 0.27 & 0.37 \\ Y_3 & -0.05 & -0.08 \\ Y_4 & 0.03 & 0.79 \\ Y_5 & 0.38 & -0.09 \\ Y_6 & 0.67 & -0.47 \\ Y_7 & -0.14 & -0.03 \end{vmatrix}.$$

- So U_1 is a feedback and autonomy variable (X_1 and X_5; task variables don't seem to be so important) while V_1 stresses satisfaction through supervisor and kind of work (Y_1 and Y_6).

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More Examples

- To augment this interpretation one can look at the correlations between the canonical variables and the original ones.
- If $\tilde{\boldsymbol{A}}$ and $\tilde{\boldsymbol{B}}$ have the coefficients $\tilde{\boldsymbol{a}}_i^T$ and $\tilde{\boldsymbol{b}}_i^T$ of $\tilde{U}_i$ and $\tilde{V}_i$ as their rows (these aren't normalized, but that doesn't matter),
- then the estimates are

$$\begin{aligned} & \text{CORR}\left[\hat{U}_i, X_j\right] = \text{CORR}\left[\tilde{U}_i, X_j\right] \\ = \quad & \text{CORR}\left[\tilde{\boldsymbol{a}}_i^T \mathbf{x}, \mathbf{e}_j^T \mathbf{x}\right] = \frac{\tilde{\boldsymbol{a}}_i^T \mathbf{S}_{11} \mathbf{e}_j}{\sqrt{\mathbf{e}_j^T \mathbf{S}_{11} \mathbf{e}_j}} \\ = \quad & \left[\left(\tilde{\boldsymbol{A}}^T \mathbf{S}_{11}\right) \left(diag\left(\mathbf{S}_{11}\right)\right)^{-1/2}\right]_{ij}; \end{aligned}$$

More Examples

- similarly

$$
\begin{aligned}
\text{CORR}\left[\hat{U}_i, Y_j\right] &= \left[\left(\tilde{\boldsymbol{A}}^T \mathbf{S}_{12}\right)\left(diag\left(\mathbf{S}_{22}\right)\right)^{-1/2}\right]_{ij}, \\
\text{CORR}\left[\hat{V}_i, X_j\right] &= \left[\left(\tilde{\boldsymbol{B}}^T \mathbf{S}_{21}\right)\left(diag\left(\mathbf{S}_{11}\right)\right)^{-1/2}\right]_{ij}, \\
\text{CORR}\left[\hat{V}_i, Y_j\right] &= \left[\left(\tilde{\boldsymbol{B}}^T \mathbf{S}_{22}\right)\left(diag\left(\mathbf{S}_{22}\right)\right)^{-1/2}\right]_{ij}.
\end{aligned}
$$

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More Examples

- For the first canonical pair this gives the correlations

$$\begin{array}{|lcc|}
 & (U_1, X_j) & (V_1, X_j) \\
X_1 & 0.83 & 0.46 \\
X_2 & 0.73 & 0.42 \\
X_3 & 0.75 & 0.61 \\
X_4 & 0.62 & 0.34 \\
X_5 & 0.86 & 0.48
\end{array}, \quad
\begin{array}{|lcc|}
 & (U_1, Y_j) & (V_1, Y_j) \\
Y_1 & 0.42 & 0.76 \\
Y_2 & 0.36 & 0.64 \\
Y_3 & 0.21 & 0.39 \\
Y_4 & 0.21 & 0.38 \\
Y_5 & 0.36 & 0.65 \\
Y_6 & 0.45 & 0.80 \\
Y_7 & 0.28 & 0.50
\end{array}.$$

- Here we see that U_1 has about the same correlation with all of the X_i (although still greatest with X_1 and X_5), perhaps contradicting the interpretation above based just on the coefficients of U_1.
- But the interpretation of V_1 as reflecting satisfaction through Y_1 and Y_6 continues to hold.

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More Examples

- Test $H_k : \rho_1^* \geq \cdots \geq \rho_k^* > 0 = \rho_{k+1}^* = \cdots = \rho_p^*$ – that only k of the canonical correlations are significant – for $k = 0$, then $k = 1$, etc.
- The p-values of each are

$$p_k = P\left(\begin{array}{c} \chi^2_{(p-k)(q-k)} > \\ -\left(n - 1 - \frac{p+q+1}{2}\right) \log \prod_{i=k+1}^{p} \left(1 - \hat{\rho}_i^{*2}\right) \end{array} \right).$$

k	χ^2	df	p_k
0	346.68	35	0.00
1	62.38	24	0.00
2	17.72	15	0.28
3	6.61	8	0.58
4	2.55	3	0.47

- Interpretation?

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Summary and Remark

- Sample Canonical Correlation
- Example
- Others